



Introduction to Cybersecurity

Ass. Prof. Mohammed Abdel Hamid

Faculty of Computers & Information

Luxor University

Computer Science Dept.

Lecture 9

Lecture Rules



Computer Malware and Classification Part I

Lecture Outline

- **Introduction**

- **Viruses**

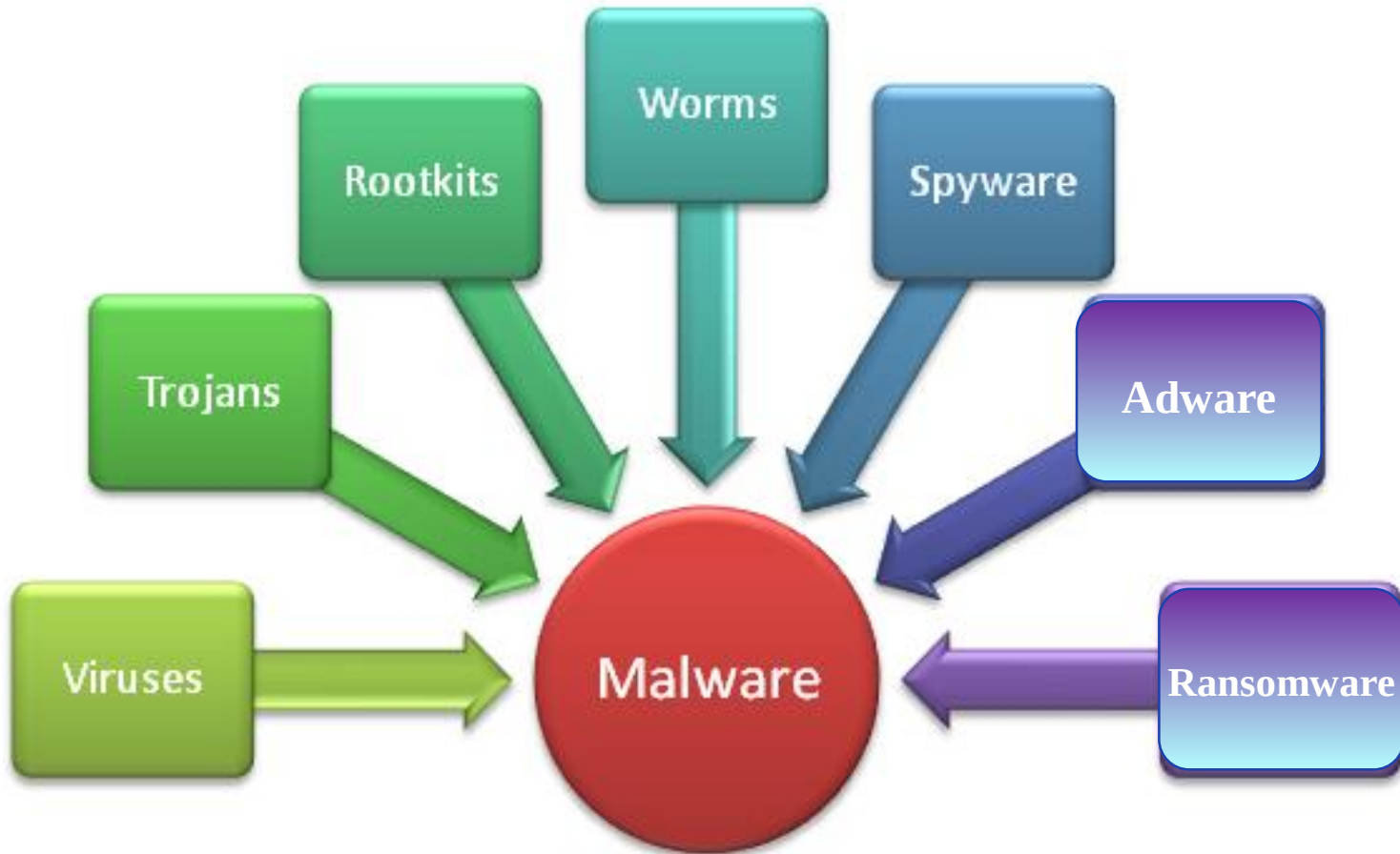
- **Trojan Horse**

- **Rootkit**

Introduction

- The **malware** is a **computer software** program **maliciously** developed to **install** on the **computers without** any **consent (permission)** of the **users**.
- The main objective is to either establish the access to the **targeted computers without** any **permission** or to **install** the **program** for creating **problems** for the **users**.
- The problems can include **data theft**, data **damage**, computer **errors**, service **stoppage**, and **spying** on the users.

Introduction

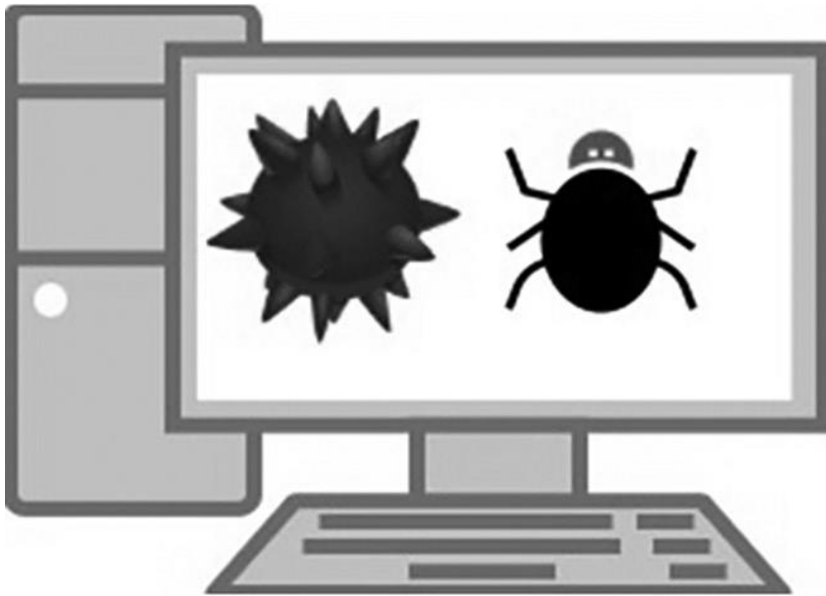


Viruses

- A **computer virus** is a **malicious computer program**, which is **designed** to **alter** the computer **functions**, **slow down** the computer **performance**, and **damage** the **valuable** files on the **computer drive**.
- The virus **programs** are **executable files** when **run** on any **machine** by **mistake**, or through any **trick** or **click**, it will get **activated** and will **start altering** the computer **configurations** and the **process**.

Viruses

- The viruses are just **programs** in the **form** of **executable** files. **They** do not get **activate** without **running** those **executable** files on your **computers**



Virus Sources

- The main **sources** of virus **propagation** include the following:
 - **Copying** on **hard drive**.
 - **Data copying** through **flash**.
 - **Email attachments**.
 - **Short** text **messages**.
 - **Fraud websites**, and **fraud social media** links.
 - **Infected** file **downloading** from **Internet**.
 - **Visiting infected websites**.

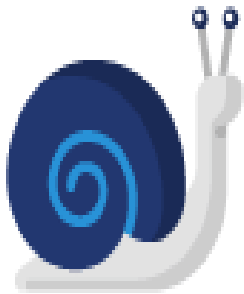
Virus Objectives

- The **objectives** of spreading a virus may include the following:
 - **Altering** computer **functioning**.
 - **Corrupting** data files on **computer**.
 - **Sending spamming** emails from your computer.
 - **Erasing valuable** data.
 - **Damaging hard** drives.

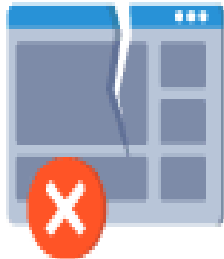
Virus Symptoms

- The main **symptoms** of a **virus-infected** computer may include the following:
 - **Frequent** appearance of **pop-ups** and other **annoyance**
 - **Changes** in the **home screen** and other **settings**
 - **Redirecting** your **online** to certain **websites**
 - **System** starts **crashing frequently**.
 - **Slowing down** of the **performance** of your **computer**.
 - **Restarting** your **computer unexpectedly**.

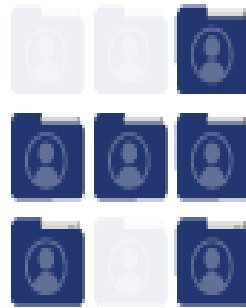
Virus Symptoms



**SLOW
PERFORMANCE**



**FREEZE-UPS
AND CRASHING**



**MISSING
FILES**



**NEW FILES
APPEAR**



**PROBLEMS
WITH
HARDWARE**



**COMPUTER
OPERATING
BY ITSELF**

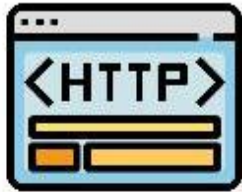
Countermeasures

- We can **save** our **computers** from the **attack** of **viruses** by taking the **following measures**:
 - **Avoid** using **insecure websites**.
 - **Do not** open **fake websites** that attract you for **free**.
 - **Always** keep your **antivirus** software **updated**.
 - Always keep **operating system (OS)** of **your** computer **updated**.
 - Turn on the **security firewall settings**.

Countermeasures Cont.

- We can save our **computers** from the attack of viruses by taking the following **measures**:
 - Always **configure** your **browser** for high **level** of **security**.
 - **Never insert flash cards** or other **storage** devices into your **computer** **without scanning** and knowing about the type of data.
 - **Never** open **emails** and **their attachments** from **unknown** people and **email address**.

Countermeasures Cont.



**Don't Click
suspicious links**



**Choose strong
password for your system**



**Keep your system
updated**



**Use Antivirus
Softwares**



**Always keep your
important files in backup**

Viruses Examples



Klez



Sasser



ILOVEYOU



Conficker



Shamoon



Nimda



Mydoom.A.exe



Mydoom.G.exe



Mydoom.M.exe



Mydoom.R.exe



Mydoom.AA.exe



Mydoom.AF.exe



Mydoom.F.exe



Mydoom.K.exe



Mydoom.O.exe



Mydoom.W.exe



Mydoom.AD.exe



Mydoom.AL.exe
Flash Player 5.0 r30
Macromedia, Inc.

Trojan Horse

- A type of **malicious software** that **disguises** itself as **legitimate software** to gain **access** to **users' systems**.
- Once inside, it can **cause** a **host** of issues, including **data theft**, **system damage**, and even **identity theft**.
- The main **objective** of **Trojan** horse are to **spy** on the **user activities** and **send** back to its **command control**.
- **Hackers** use **Trojan** to open the **backdoor** to **access** your computer and **establish control** on **your system**.

Trojan Horse Cont.

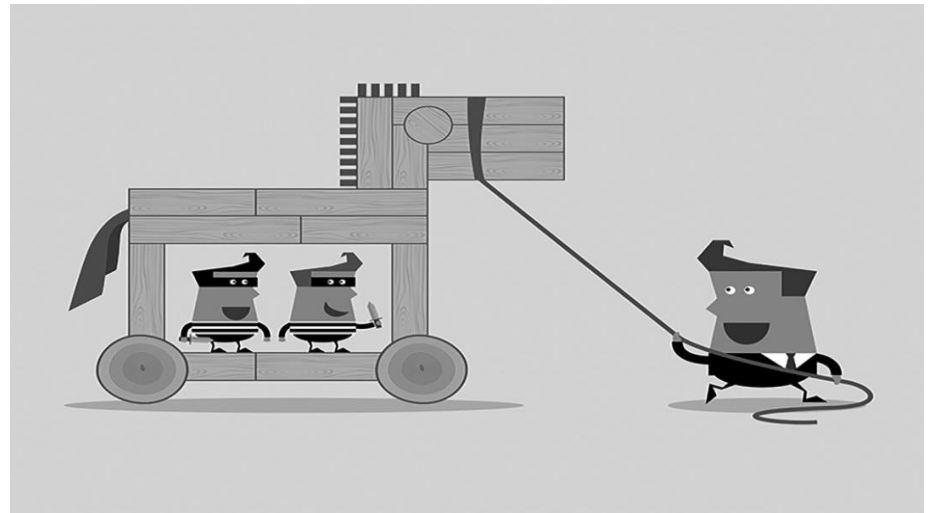
- According to the **statistics**, in the year **2016**, Trojan **horse-based** cyber **infections** accounted for about **74.99%**.
- The **Trojan horses cannot replicate** as the **virus** or **computer worms** can do.
- **Trojan dropper** is the second **biggest** threat in **2018**, according to the **Kaspersky Lab research**.
- **Trojan horses** may look **less dangerous** on **stand alone** systems, but in fact, they are very **dangerous** in an **online business environment**.

Trojan Horse Objectives

- **Objectives** of a **Trojan horse** on an **infected** computer include the **following**:
 - **Collecting** data and **sending** to **command** and **control** center.
 - **Copying** the **files** and **credential information**.
 - **Blocking** of the **data**.
 - **Altering** the **useful** data.
 - **Reducing** the **performance** of computer.
 - **Deleting** some data files.

Trojan Horse Types

- There are **numerous** types of **Trojan horses** used in the **modern cybercriminal** activities as follows:
- Trojan Spy
- Trojan Dropper
- Trojan DDoS
- Trojan Mail Finder
- Trojan Proxy
- Trojan Clicker
- Trojan Ransom
- Trojan SMS



Modern Trojan Horse Types



Backdoor Trojans



Fraudulent antivirus Trojans



Downloader Trojans



Mailfinder Trojans



Rootkit Trojans



Remote Access Trojans

Trojan Horse Symptoms

- The **main symptoms** of a **Trojan horse** infected computer may include the following:
 - **Performance** degrades **significantly**.
 - **Internet** speed **slows down**.
 - **Problem** in Internet **browsing occurs**, **pop-ups** may **appear**.
 - **Large** number of **spam emails** appear in **inbox**.
 - **Your** contacts may **receive** emails that you **did not send** to them.
 - **Your** data files are **deleted** or **modified**.

Trojan Horse Detection Example



Scan Complete!
8 Security Issues Found

8 / 89922



BUY!

V: 5.11.8
DB: 2021.11.22x01



Home



Active Scan



Malware/PC Scan



System Guard



Settings



HelpDesk



About

☒ Malware: 8

Below is a list of malware objects detected on your computer. Malware may include spyware, adware, trojans, ransomware, worms, viruses, and rootkits. Malware generally represents a security threat and should be removed from your system as soon as possible. Each of the objects identified below has been pre-set to be removed. Should you wish not to remove any of these objects, you may uncheck them below.

☐ ☒ **Severity Level: 90% (High)** Trojan.Dropper (8 objects)

Dropper is a Trojan that stealth installs itself, most often by exploiting system vulnerabilities and security flaws, through software downloads from malicious websites or via email attachments from untrusted sources (usually .exe, .pif, .avi, and even .jpg files). Dropper runs silently in the background and performs malicious functions. Dropper may allow hackers to gain complete control over your system after installation. Dropper may covertly delete files, install additional malware, steal passwords and banking information, modify your system security settings, and monitor and transmit your computer activity and other highly sensitive personal data to a remote attacker.

☐ ☒ [F] %LOCALAPPDATA%\SystemDir\nethost.exe

☐ ☒ [F] %WINDIR%\System32\regsvr.exe

☐ ☒ [F] %WINDIR%\GoogleToolbarUpdate.exe

☐ ☒ [F] %WINDIR%\system32\explorer\winlogon.exe

☐ ☒ [F] %WINDIR%\system32\evans.exe

☐ ☒ [F] %TEMP%\scvhost.exe

☐ ☒ [F] drm.exe

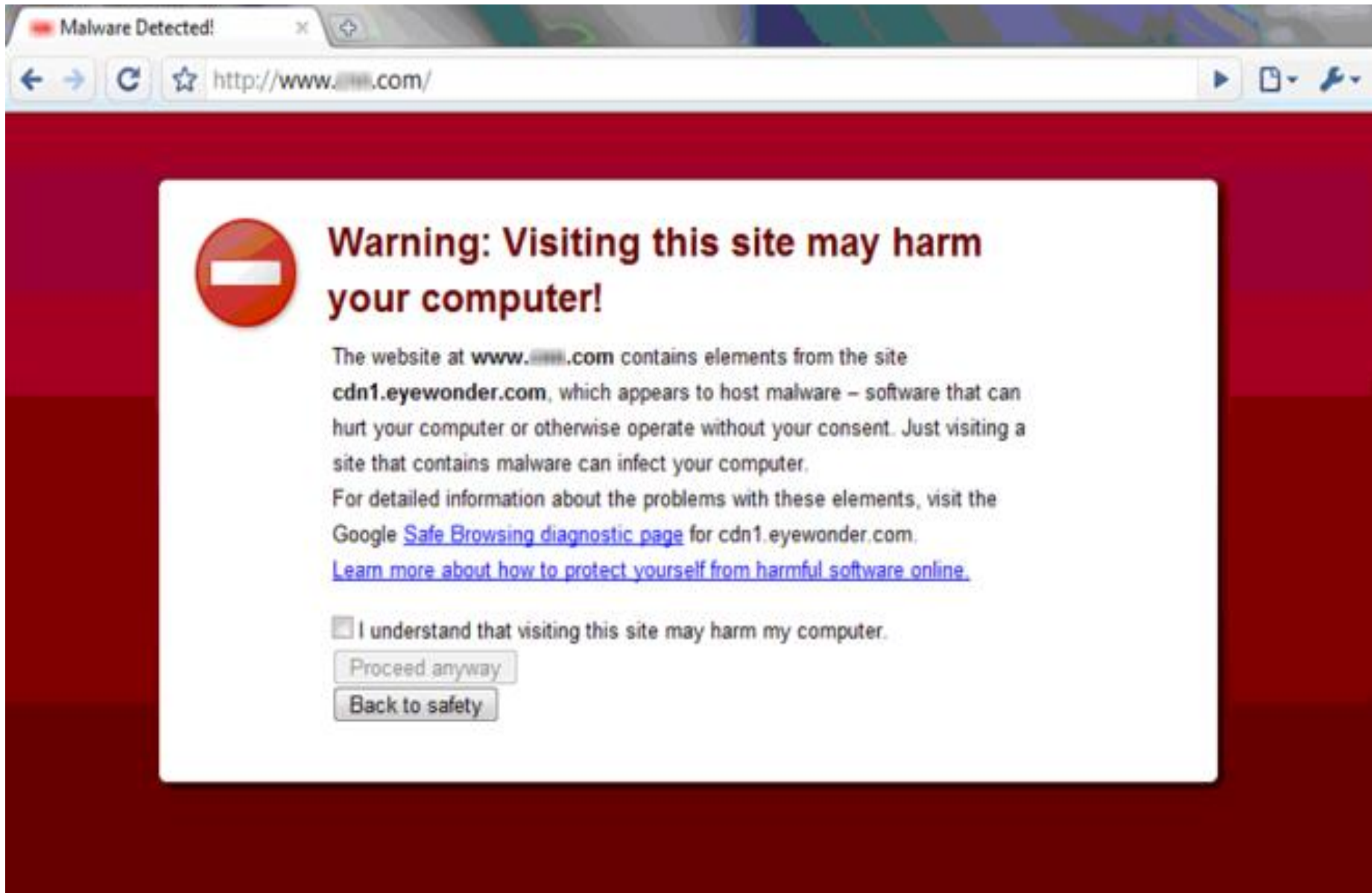


EnigmaSoft

Next

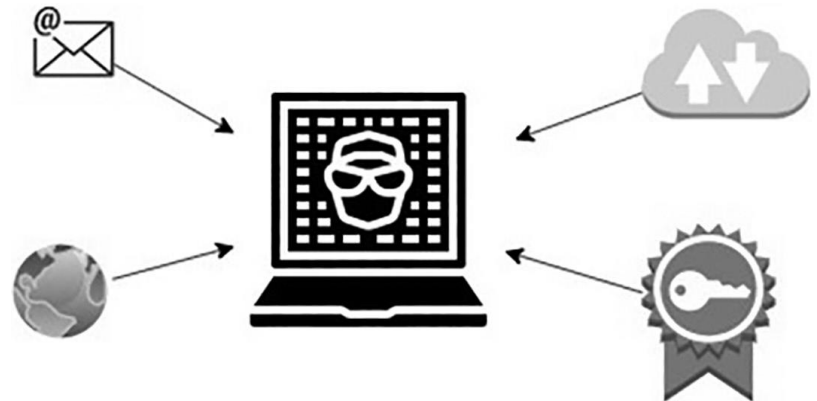


Trojan Horse Detection Example



Rootkit

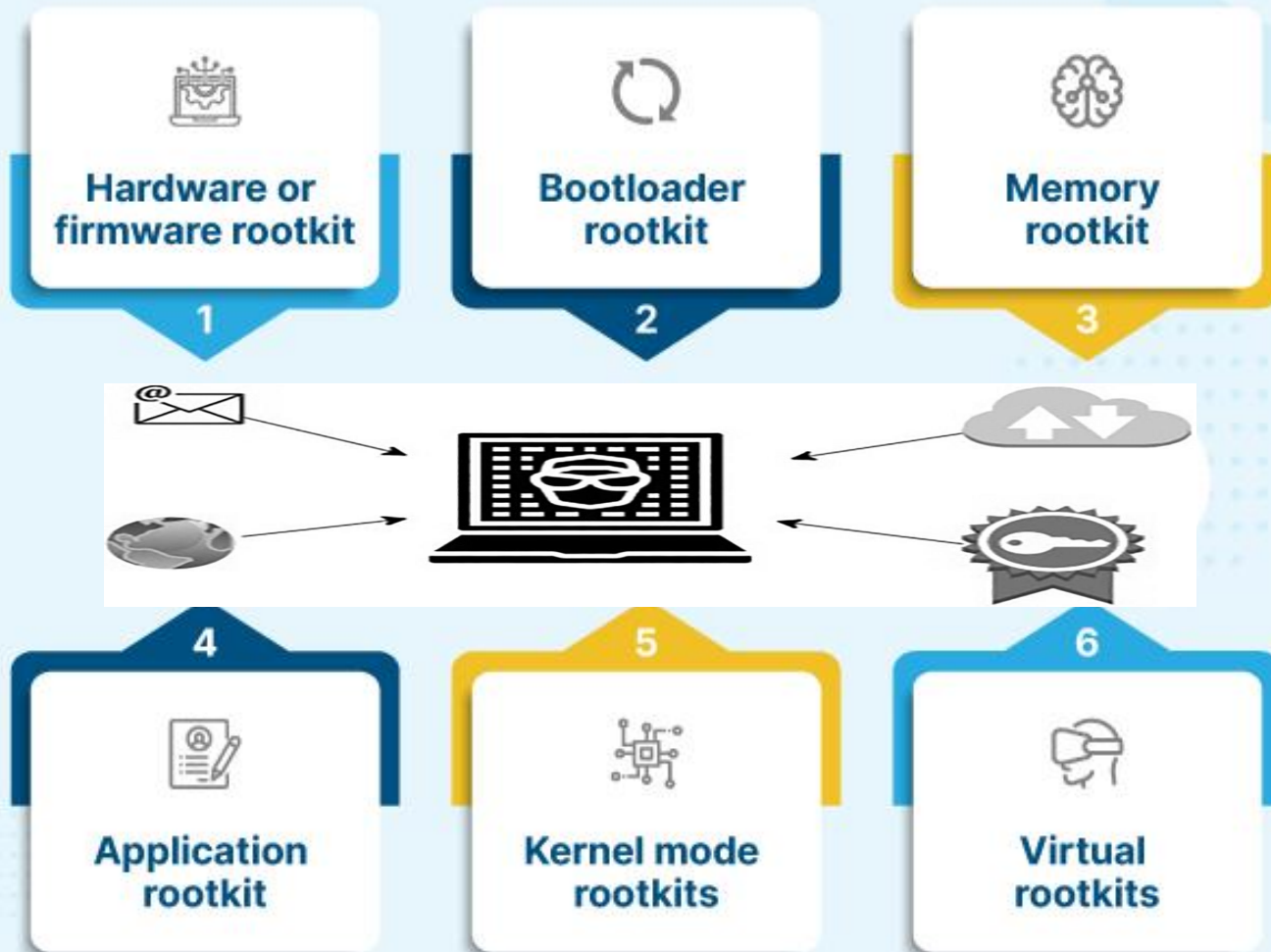
- **Rootkit** is a type of **malware** that gets **administrator-level privileges** on the **OS** of the **computer without** showing its **presence** on the **computer**.
- The **main feature** of **rootkit** is that it **hides** from being **detected easily**, but **maintains** the **control** over the **OS** to **perform** its **critical tasks** on the **system**.



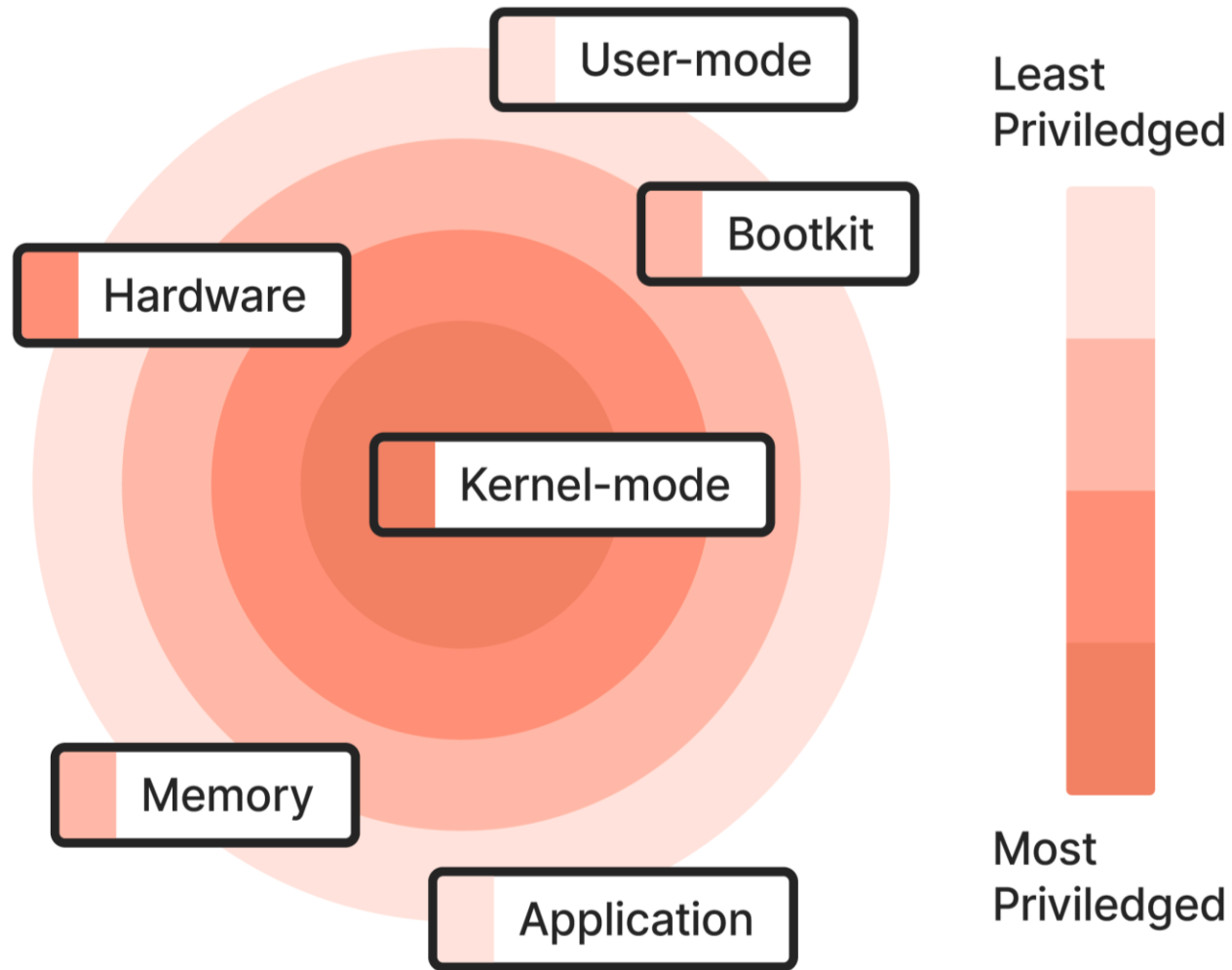
How it Works?

- The rootkit **malware prevent** the **request** sent to the **OS** from **reaching** to the **desired API** and **responds** with **fake responses**.
- Moreover, you **request** the **OS** to **start** the **antivirus** or **anti-malware** application to **scan** your **computer**; the **request** is **interrupted from** reaching to **the OS**.

Rootkit Types



Rootkit Types



Rootkit Symptoms

- If the computer shows the following **symptoms**, then you should suspect that the computer has been **infected** by the **rootkit** attack.
 - **Modifications** in the dates and time of computer
 - **Slowing** down of the **computer**
 - **Unexpected** system **error messages**
 - **Many programs**, especially **computer security-related** programs, **fail** to **start**
 - **Appearance** of **blue screen** of **death**

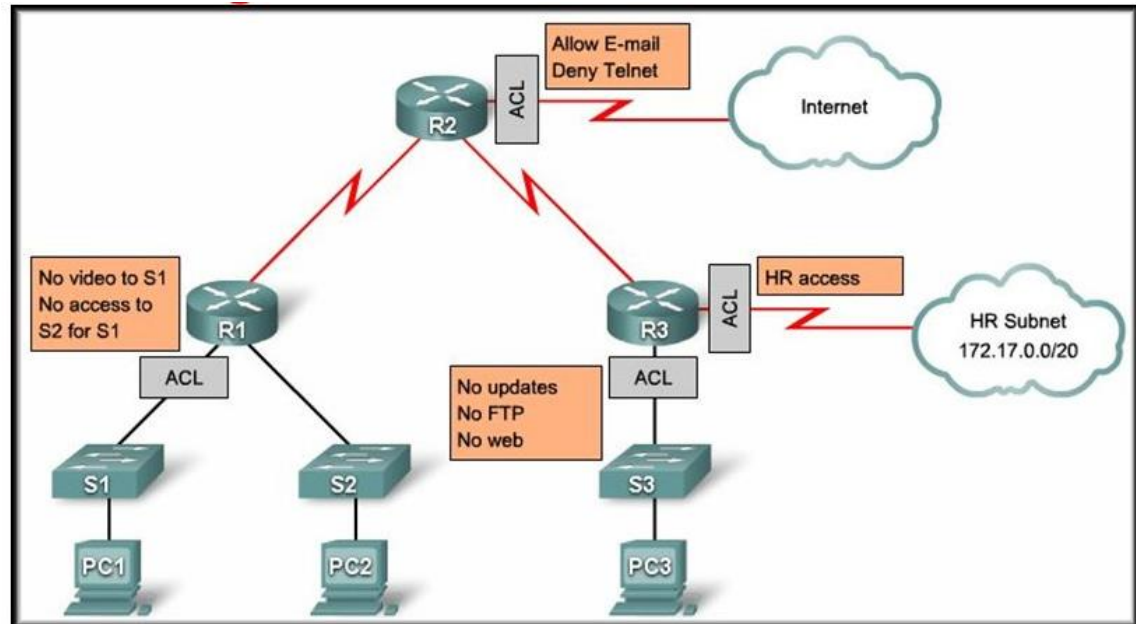
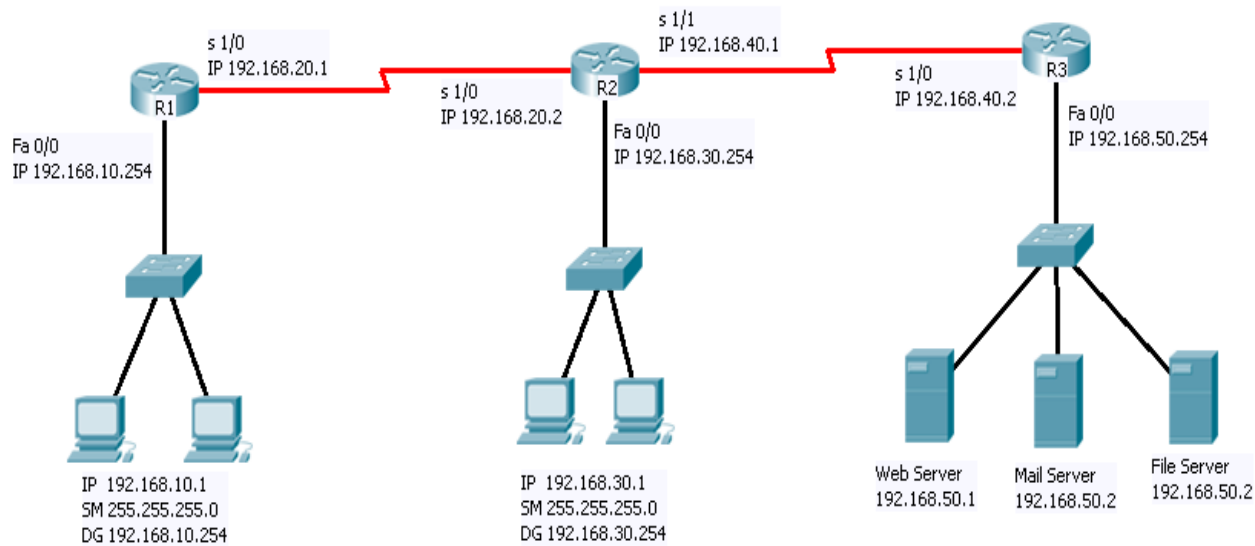
Countermeasures

- We can **save** our **computers** from the **attack** of **rootkit** by taking the **following measures**:
 - **Always** visit the **legitimate** and **secure websites**.
 - Do not **download free products**.
 - **Open emails** and their **attachments** from **known people only**.
 - Never **open attachments** in **SMS** and other **messages**.

Countermeasures Cont.

- We can **save** our **computers** from the **attack** of **rootkit** by taking the **following measures**:
 - **Activate** access control **lists** (**ACLs**) on the **system** and **network**.
 - **Keep** your **anti-malware** up to date.
 - **Keep OS** up to date.
 - **Activate** firewall on **your system**.
 - **Activate** intrusion prevention system (**IPS**).
 - Use the **latest anti-rootkit** tools with **Security Technology and Response (STAR)**.

ACLs in Cisco



The screenshot shows the 'Status' window of Symantec Endpoint Protection. At the top, a yellow banner indicates a 'Security compliance scan failed.' with a message: 'Your computer failed its security compliance check. Please see the Client Management Security log for more information.' Below this, a list of installed components is shown:

- Virus and Spyware Protection**: Protects against viruses, malware, and spyware. Definitions: 24 March 2012 r19. An 'Options' button is visible.
- Proactive Threat Protection**: Provides zero-day protection against unknown threats. Definitions: 17 March 2012 r11. An 'Options' button is visible and highlighted with a red box.
- Network Threat Protection**: Protects against network threats. Definitions: 17 March 2012 r11.
- Network Advanced Protection**: Monitors and protects against advanced threats. Definitions: 17 March 2012 r11.

A red arrow points from the 'Options' button of 'Proactive Threat Protection' to the 'Proactive Threat Protection Logs' window. This window shows a table of threat logs:

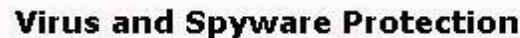
Filename	Risk	Action	Risk Type
cloudcar.exe	Bloodhound.Sonar.9	Quarantined	Application
cloudcar.exe	Bloodhound.Sonar.9	Quarantined	Application
socar.exe	Bloodhound.Sonar.9	Quarantined	Application

Help



Your computer failed its security compliance check. Please see the Client Management Security log for more information.

The following Symantec Security components are installed on your computer:



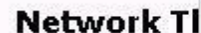
Protects against viruses, malware, and spyware
Definitions: **24 March 2012 r19**

Options



Provides zero-day protection against unknown threats
Definitions: **17 March 2012 r11**

Options



Protects again
Definitions:



Monitors and e
Definitions:

Proactive Threat Protection Logs

Threat Log | System Log

[illegible]

Questions

Q1. What is a computer virus?

Q2. State the main sources of virus propagation?

Q3. What are the types of Rootkit?

Q4. List the symptoms of Trojan horse?

Answers

A1:

- The virus programs are executable files when run on any machine by mistake, or through any trick or click, it will get activated and will start altering the computer configurations and the process, which are used for smooth operation of the computer.

Answers

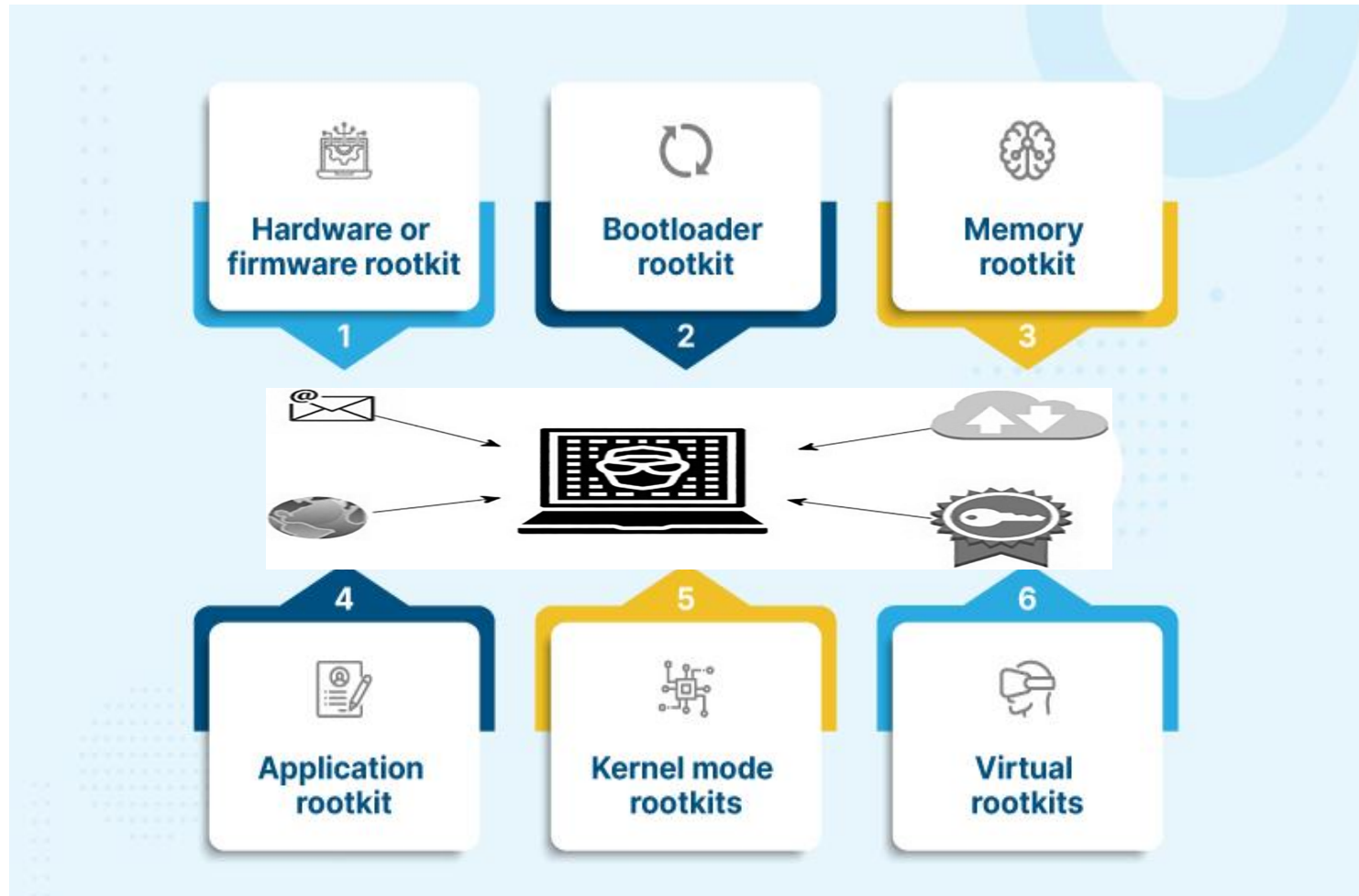
A2:

The main sources of virus propagation include the following:

- Copying on hard drive
- Data copying through flash
- Email attachments
- Short text messages
- Scam websites
- Scam social media links
- Infected file downloading from Internet
- Visiting infected websites

Answers

A3:



Answers

A4:

The **main symptoms** of a **Trojan horse** infected computer may include the following:

- **Performance** degrades **significantly**.
- **Internet** speed **slows down**.
- **Problem** in Internet **browsing occurs**, **pop-ups** may appear.
- **Large** number of **spam emails** appear in **inbox**.
- **Your** contacts may **receive** emails that you **did not send** to them.
- **Loading** of **computer** takes **much longer**.
- **Your** data files are **deleted** or **modified**.

Task (15 Min)

1. Encrypt the following **Plaintext** using **Vigenère Cipher** where $m=5$, $k=(2,12,5,17,9)$.

- Today, Attack Security System.

Task (15 Min)

1. Decipher the following **Ciphertext** using **Vigenère Cipher** where **key = BOXENTRIQ**

- JAMVB OVGEV FMYMS CMIPZ SMAZJ SYMZP

thanks for
listening!